Python 코드 샘플

ONTAP Select

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Python 코드 샘플

클러스터를 생성하는 스크립트입니다

다음 스크립트를 사용하여 스크립트 및 JSON 입력 파일 내에 정의된 매개 변수를 기반으로 클러스터를 생성할 수 있습니다.

```
#!/usr/bin/env python
##-------------------------------------------------------------------
#
# File: cluster.py
#
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#
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#
##-------------------------------------------------------------------

import traceback
import argparse
import json
import logging

from deploy_requests import DeployRequests


def add_vcenter_credentials(deploy, config):
    """ Add credentials for the vcenter if present in the config """
    log_debug_trace()

    vcenter = config.get('vcenter', None)
    if vcenter and not deploy.resource_exists('/security/credentials',
                                              'hostname', vcenter
['hostname']):
        log_info("Registering vcenter {} credentials".format(vcenter
['hostname']))
        data = {k: vcenter[k] for k in ['hostname', 'username',
```

```
'password']}
        data['type'] = "vcenter"
        deploy.post('/security/credentials', data)


def add_standalone_host_credentials(deploy, config):
    """ Add credentials for standalone hosts if present in the config.
        Does nothing if the host credential already exists on the Deploy.
    """
    log_debug_trace()

    hosts = config.get('hosts', [])
    for host in hosts:
        # The presense of the 'password' will be used only for standalone
hosts.
        # If this host is managed by a vcenter, it should not have a host
'password' in the json.
        if 'password' in host and not deploy.resource_exists
('/security/credentials',
                                                                 'hostname',
host['name']):
            log_info("Registering host {} credentials".format(host[
'name']))
            data = {'hostname': host['name'], 'type': 'host',
                    'username': host['username'], 'password': host
['password']}
            deploy.post('/security/credentials', data)


def register_unkown_hosts(deploy, config):
    ''' Registers all hosts with the deploy server.
        The host details are read from the cluster config json file.

        This method will skip any hosts that are already registered.
        This method will exit the script if no hosts are found in the
config.
    '''
    log_debug_trace()

    data = {"hosts": []}
    if 'hosts' not in config or not config['hosts']:
        log_and_exit("The cluster config requires at least 1 entry in the
'hosts' list got {}".format(config))

    missing_host_cnt = 0
    for host in config['hosts']:
```

```
        if not deploy.resource_exists('/hosts', 'name', host['name']):
            missing_host_cnt += 1
            host_config = {"name": host['name'], "hypervisor_type": host
['type']}
            if 'mgmt_server' in host:
                host_config["management_server"] = host['mgmt_server']
                log_info(
                   "Registering from vcenter {mgmt_server}".format(**
host))

            if 'password' in host and 'user' in host:
                host_config['credential'] = {
                    "password": host['password'], "username": host[
'user']}

            log_info("Registering {type} host {name}".format(**host))
            data["hosts"].append(host_config)

    # only post /hosts if some missing hosts were found
    if missing_host_cnt:
        deploy.post('/hosts', data, wait_for_job=True)


def add_cluster_attributes(deploy, config):
    ''' POST a new cluster with all needed attribute values.
        Returns the cluster_id of the new config
    '''
    log_debug_trace()

    cluster_config = config['cluster']
    cluster_id = deploy.find_resource('/clusters', 'name', cluster_config
['name'])

    if not cluster_id:
        log_info("Creating cluster config named {name}".format(
**cluster_config))

        # Filter to only the valid attributes, ignores anything else in
the json
        data = {k: cluster_config[k] for k in [
            'name', 'ip', 'gateway', 'netmask', 'ontap_image_version',
'dns_info', 'ntp_servers']}

        num_nodes = len(config['nodes'])

        log_info("Cluster properties: {}".format(data))
```

```
        resp = deploy.post('/v3/clusters?node_count={}'.format(num_nodes),
data)
        cluster_id = resp.headers.get('Location').split('/')[-1]

    return cluster_id


def get_node_ids(deploy, cluster_id):
    ''' Get the the ids of the nodes in a cluster. Returns a list of
node_ids.'''
    log_debug_trace()

    response = deploy.get('/clusters/{}/nodes'.format(cluster_id))
    node_ids = [node['id'] for node in response.json().get('records')]
    return node_ids


def add_node_attributes(deploy, cluster_id, node_id, node):
    ''' Set all the needed properties on a node '''
    log_debug_trace()

    log_info("Adding node '{}' properties".format(node_id))

    data = {k: node[k] for k in ['ip', 'serial_number', 'instance_type',
                                 'is_storage_efficiency_enabled'] if k in
node}
    # Optional: Set a serial_number
    if 'license' in node:
        data['license'] = {'id': node['license']}

    # Assign the host
    host_id = deploy.find_resource('/hosts', 'name', node['host_name'])
    if not host_id:
        log_and_exit("Host names must match in the 'hosts' array, and the
nodes.host_name property")

    data['host'] = {'id': host_id}

    # Set the correct raid_type
    is_hw_raid = not node['storage'].get('disks')  # The presence of a
list of disks indicates sw_raid
    data['passthrough_disks'] = not is_hw_raid

    # Optionally set a custom node name
    if 'name' in node:
        data['name'] = node['name']
```

```
    log_info("Node properties: {}".format(data))
    deploy.patch('/clusters/{}/nodes/{}'.format(cluster_id, node_id),
data)


def add_node_networks(deploy, cluster_id, node_id, node):
    ''' Set the network information for a node '''
    log_debug_trace()

    log_info("Adding node '{}' network properties".format(node_id))

    num_nodes = deploy.get_num_records('/clusters/{}/nodes'.format
(cluster_id))

    for network in node['networks']:

        # single node clusters do not use the 'internal' network
        if num_nodes == 1 and network['purpose'] == 'internal':
            continue

        # Deduce the network id given the purpose for each entry
        network_id = deploy.find_resource(
'/clusters/{}/nodes/{}/networks'.format(cluster_id, node_id),
                                          'purpose', network['purpose'])
        data = {"name": network['name']}
        if 'vlan' in network and network['vlan']:
            data['vlan_id'] = network['vlan']

        deploy.patch('/clusters/{}/nodes/{}/networks/{}'.format(
cluster_id, node_id, network_id), data)


def add_node_storage(deploy, cluster_id, node_id, node):
    ''' Set all the storage information on a node '''
    log_debug_trace()

    log_info("Adding node '{}' storage properties".format(node_id))
    log_info("Node storage: {}".format(node['storage']['pools']))

    data = {'pool_array': node['storage']['pools']}  # use all the json
properties
    deploy.post(
        '/clusters/{}/nodes/{}/storage/pools'.format(cluster_id, node_id),
data)

    if 'disks' in node['storage'] and node['storage']['disks']:
        data = {'disks': node['storage']['disks']}
```

```
        deploy.post(
            '/clusters/{}/nodes/{}/storage/disks'.format(cluster_id,
node_id), data)


def create_cluster_config(deploy, config):
    ''' Construct a cluster config in the deploy server using the input
json data '''
    log_debug_trace()

    cluster_id = add_cluster_attributes(deploy, config)

    node_ids = get_node_ids(deploy, cluster_id)
    node_configs = config['nodes']

    for node_id, node_config in zip(node_ids, node_configs):
        add_node_attributes(deploy, cluster_id, node_id, node_config)
        add_node_networks(deploy, cluster_id, node_id, node_config)
        add_node_storage(deploy, cluster_id, node_id, node_config)

    return cluster_id


def deploy_cluster(deploy, cluster_id, config):
    ''' Deploy the cluster config to create the ONTAP Select VMs. '''
    log_debug_trace()
    log_info("Deploying cluster: {}".format(cluster_id))

    data = {'ontap_credential': {'password': config['cluster'
]['ontap_admin_password']}}
    deploy.post('/clusters/{}/deploy?inhibit_rollback=true'.format
(cluster_id),
                data, wait_for_job=True)


def log_debug_trace():
    stack = traceback.extract_stack()
    parent_function = stack[-2][2]
    logging.getLogger('deploy').debug('Calling %s()' % parent_function)


def log_info(msg):
    logging.getLogger('deploy').info(msg)


def log_and_exit(msg):
    logging.getLogger('deploy').error(msg)
```

```
    exit(1)


def configure_logging(verbose):
    FORMAT = '%(asctime)-15s:%(levelname)s:%(name)s: %(message)s'
    if verbose:
        logging.basicConfig(level=logging.DEBUG, format=FORMAT)
    else:
        logging.basicConfig(level=logging.INFO, format=FORMAT)
        logging.getLogger('requests.packages.urllib3.connectionpool'
).setLevel(
            logging.WARNING)


def main(args):
    configure_logging(args.verbose)
    deploy = DeployRequests(args.deploy, args.password)

    with open(args.config_file) as json_data:
        config = json.load(json_data)

        add_vcenter_credentials(deploy, config)

        add_standalone_host_credentials(deploy, config)

        register_unkown_hosts(deploy, config)

        cluster_id = create_cluster_config(deploy, config)

        deploy_cluster(deploy, cluster_id, config)


def parseArgs():
    parser = argparse.ArgumentParser(description='Uses the ONTAP Select
Deploy API to construct and deploy a cluster.')
    parser.add_argument('-d', '--deploy', help='Hostname or IP address of
Deploy server')
    parser.add_argument('-p', '--password', help='Admin password of Deploy
server')
    parser.add_argument('-c', '--config_file', help='Filename of the
cluster config')
    parser.add_argument('-v', '--verbose', help='Display extra debugging
messages for seeing exact API calls and responses',
                        action='store_true', default=False)
    return parser.parse_args()

if __name__ == '__main__':
```

```
    args = parseArgs()
    main(args)
```

따라서 스크립트를 사용하여 클러스터를 생성하는 JSON입니다

Python 코드 샘플을 사용하여 ONTAP Select 클러스터를 생성하거나 삭제할 때 JSON 파일을 스크립트에 입력으로 제공해야 합니다. 배포 계획에 따라 적절한 JSON 샘플을 복사하여 수정할 수 있습니다.

ESXi의 단일 노드 클러스터

```
{
  "hosts": [
    {
      "password": "mypassword1",
      "name": "host-1234",
      "type": "ESX",
      "username": "admin"
    }
  ],

  "cluster": {
    "dns_info": {
      "domains": ["lab1.company-demo.com", "lab2.company-demo.com",
        "lab3.company-demo.com", "lab4.company-demo.com"
        ],

      "dns_ips": ["10.206.80.135", "10.206.80.136"]
      },
      "ontap_image_version": "9.7",
      "gateway": "10.206.80.1",
      "ip": "10.206.80.115",
      "name": "mycluster",
      "ntp_servers": ["10.206.80.183", "10.206.80.142"],
      "ontap_admin_password": "mypassword2",
      "netmask": "255.255.254.0"
  },

  "nodes": [
    {
      "serial_number": "3200000nn",
      "ip": "10.206.80.114",
      "name": "node-1",
      "networks": [
```

```
        {
          "name": "ontap-external",
          "purpose": "mgmt",
          "vlan": 1234
        },
        {
          "name": "ontap-external",
          "purpose": "data",
          "vlan": null
        },
        {
          "name": "ontap-internal",
          "purpose": "internal",
          "vlan": null
        }
      ],
      "host_name": "host-1234",
      "is_storage_efficiency_enabled": false,
      "instance_type": "small",
      "storage": {
        "disk": [],
        "pools": [
          {
            "name": "storage-pool-1",
            "capacity": 4802666790125
          }
        ]
      }
    }
  ]
}
```


vCenter를 사용하는 ESXi의 단일 노드 클러스터


```
{
  "hosts": [
    {
      "name":"host-1234",
      "type":"ESX",
      "mgmt_server":"vcenter-1234"
    }
  ],

  "cluster": {
    "dns_info": {"domains": ["lab1.company-demo.com", "lab2.company-
```

```
demo.com",
      "lab3.company-demo.com", "lab4.company-demo.com"
      ],
      "dns_ips": ["10.206.80.135","10.206.80.136"]
  },

  "ontap_image_version":"9.7",
  "gateway":"10.206.80.1",
  "ip":"10.206.80.115",
  "name":"mycluster",
  "ntp_servers": ["10.206.80.183","10.206.80.142"],
  "ontap_admin_password":"mypassword2",
  "netmask":"255.255.254.0"
  },

  "vcenter": {
    "password":"mypassword2",
    "hostname":"vcenter-1234",
    "username":"selectadmin"
  },

  "nodes": [
    {
      "serial_number": "3200000nn",
      "ip":"10.206.80.114",
      "name":"node-1",
      "networks": [
        {
          "name":"ONTAP-Management",
          "purpose":"mgmt",
          "vlan":null
        },
        {
          "name": "ONTAP-External",
          "purpose":"data",
          "vlan":null
        },
        {
          "name": "ONTAP-Internal",
          "purpose":"internal",
          "vlan":null
        }
      ],

      "host_name": "host-1234",
      "is_storage_efficiency_enabled": false,
```

```
      "instance_type": "small",
      "storage": {
        "disk":[],
        "pools": [
          {
            "name": "storage-pool-1",
            "capacity":5685190380748
          }
        ]
      }
    }
  ]
}
```

KVM의 단일 노드 클러스터

- ONTAP Select 9.10.1부터는 KVM 하이퍼바이저에 새 클러스터를 더 이상 구축할 수 없습니다.
- ONTAP Select 9.11.1부터는 오프라인 및 삭제 기능을 제외한 모든 관리 기능을 기존 KVM 클러스터와 호스트에서 사용할 수 없습니다.

```
{
  "hosts": [
    {
      "password": "mypassword1",
      "name":"host-1234",
      "type":"KVM",
      "username":"root"
    }
  ],

  "cluster": {
    "dns_info": {
      "domains": ["lab1.company-demo.com", "lab2.company-demo.com",
        "lab3.company-demo.com", "lab4.company-demo.com"
      ],

      "dns_ips": ["10.206.80.135", "10.206.80.136"]
    },

    "ontap_image_version": "9.7",
    "gateway":"10.206.80.1",
    "ip":"10.206.80.115",
    "name":"CBF4ED97",
    "ntp_servers": ["10.206.80.183", "10.206.80.142"],
```

```
    "ontap_admin_password": "mypassword2",
    "netmask":"255.255.254.0"
  },
  "nodes": [
    {
      "serial_number":"3200000nn",
      "ip":"10.206.80.115",
      "name": "node-1",
      "networks": [
        {
          "name": "ontap-external",
          "purpose": "mgmt",
          "vlan":1234
        },
        {
          "name": "ontap-external",
          "purpose": "data",
          "vlan": null
        },
        {
          "name": "ontap-internal",
          "purpose": "internal",
          "vlan": null
        }
      ],

      "host_name": "host-1234",
      "is_storage_efficiency_enabled": false,
      "instance_type": "small",
      "storage": {
        "disk": [],
        "pools": [
          {
            "name": "storage-pool-1",
            "capacity": 4802666790125
          }
        ]
      }
    }
  ]
}
```

노드 라이센스를 추가하는 스크립트입니다

다음 스크립트를 사용하여 ONTAP Select 노드에 대한 라이센스를 추가할 수 있습니다.

```python
#!/usr/bin/env python
##-----------------------------------------------------------------------
#
# File: add_license.py
#
# (C) Copyright 2019 NetApp, Inc.
#
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# any kind, including but not limited for warranties of merchantability
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# no less restrictive than those set forth herein.
#
##-----------------------------------------------------------------------

import argparse
import logging
import json

from deploy_requests import DeployRequests


def post_new_license(deploy, license_filename):
    log_info('Posting a new license: {}'.format(license_filename))

    # Stream the file as multipart/form-data
    deploy.post('/licensing/licenses', data={},
                files={'license_file': open(license_filename, 'rb')})

    # Alternative if the NLF license data is converted to a string.
    # with open(license_filename, 'rb') as f:
    #    nlf_data = f.read()
    #    r = deploy.post('/licensing/licenses', data={},
    #                    files={'license_file': (license_filename,
nlf_data)})


def put_license(deploy, serial_number, data, files):
    log_info('Adding license for serial number: {}'.format(serial_number))

    deploy.put('/licensing/licenses/{}'.format(serial_number), data=data,
files=files)
```

```
def put_used_license(deploy, serial_number, license_filename,
ontap_username, ontap_password):
    ''' If the license is used by an 'online' cluster, a username/password
must be given. '''

    data = {'ontap_username': ontap_username, 'ontap_password':
ontap_password}
    files = {'license_file': open(license_filename, 'rb')}

    put_license(deploy, serial_number, data, files)


def put_free_license(deploy, serial_number, license_filename):
    data = {}
    files = {'license_file': open(license_filename, 'rb')}

    put_license(deploy, serial_number, data, files)


def get_serial_number_from_license(license_filename):
    ''' Read the NLF file to extract the serial number '''
    with open(license_filename) as f:
        data = json.load(f)

        statusResp = data.get('statusResp', {})
        serialNumber = statusResp.get('serialNumber')
        if not serialNumber:
            log_and_exit("The license file seems to be missing the
serialNumber")

        return serialNumber


def log_info(msg):
    logging.getLogger('deploy').info(msg)


def log_and_exit(msg):
    logging.getLogger('deploy').error(msg)
    exit(1)


def configure_logging():
    FORMAT = '%(asctime)-15s:%(levelname)s:%(name)s: %(message)s'
    logging.basicConfig(level=logging.INFO, format=FORMAT)
```

```
    logging.getLogger('requests.packages.urllib3.connectionpool').
setLevel(logging.WARNING)


def main(args):
    configure_logging()
    serial_number = get_serial_number_from_license(args.license)

    deploy = DeployRequests(args.deploy, args.password)

    # First check if there is already a license resource for this serial-
number
    if deploy.find_resource('/licensing/licenses', 'id', serial_number):

        # If the license already exists in the Deploy server, determine if
its used
        if deploy.find_resource('/clusters', 'nodes.serial_number',
serial_number):

            # In this case, requires ONTAP creds to push the license to
the node
            if args.ontap_username and args.ontap_password:
                put_used_license(deploy, serial_number, args.license,
                                 args.ontap_username, args.ontap_password)
            else:
                print "ERROR: The serial number for this license is in
use. Please provide ONTAP credentials."
        else:
            # License exists, but its not used
            put_free_license(deploy, serial_number, args.license)
    else:
        # No license exists, so register a new one as an available license
for later use
        post_new_license(deploy, args.license)


def parseArgs():
    parser = argparse.ArgumentParser(description='Uses the ONTAP Select
Deploy API to add or update a new or used NLF license file.')
    parser.add_argument('-d', '--deploy', required=True, type=str, help
='Hostname or IP address of ONTAP Select Deploy')
    parser.add_argument('-p', '--password', required=True, type=str, help
='Admin password of Deploy server')
    parser.add_argument('-l', '--license', required=True, type=str, help
='Filename of the NLF license data')
    parser.add_argument('-u', '--ontap_username', type=str,
```

```
                        help='ONTAP Select username with privelege to add
the license. Only provide if the license is used by a Node.')
    parser.add_argument('-o', '--ontap_password', type=str,
                        help='ONTAP Select password for the
ontap_username. Required only if ontap_username is given.')
    return parser.parse_args()

if __name__ == '__main__':
    args = parseArgs()
    main(args)
```

클러스터를 삭제하는 스크립트입니다

다음 CLI 스크립트를 사용하여 기존 클러스터를 삭제할 수 있습니다.

```
#!/usr/bin/env python
##-------------------------------------------------------------------
#
# File: delete_cluster.py
#
# (C) Copyright 2019 NetApp, Inc.
#
# This sample code is provided AS IS, with no support or warranties of
# any kind, including but not limited for warranties of merchantability
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# solely for the purpose of researching, designing, developing and
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# provided that the above copyright notice appears in all copies and
# that the software application product is distributed pursuant to terms
# no less restrictive than those set forth herein.
#
##-------------------------------------------------------------------

import argparse
import json
import logging

from deploy_requests import DeployRequests

def find_cluster(deploy, cluster_name):
    return deploy.find_resource('/clusters', 'name', cluster_name)


def offline_cluster(deploy, cluster_id):
```

```
    # Test that the cluster is online, otherwise do nothing
    response = deploy.get('/clusters/{}?fields=state'.format(cluster_id))
    cluster_data = response.json()['record']
    if cluster_data['state'] == 'powered_on':
        log_info("Found the cluster to be online, modifying it to be
powered_off.")
        deploy.patch('/clusters/{}'.format(cluster_id), {'availability':
'powered_off'}, True)


def delete_cluster(deploy, cluster_id):
    log_info("Deleting the cluster({}).".format(cluster_id))
    deploy.delete('/clusters/{}'.format(cluster_id), True)
    pass


def log_info(msg):
    logging.getLogger('deploy').info(msg)


def configure_logging():
    FORMAT = '%(asctime)-15s:%(levelname)s:%(name)s: %(message)s'
    logging.basicConfig(level=logging.INFO, format=FORMAT)
    logging.getLogger('requests.packages.urllib3.connectionpool').
setLevel(logging.WARNING)


def main(args):
    configure_logging()
    deploy = DeployRequests(args.deploy, args.password)

    with open(args.config_file) as json_data:
        config = json.load(json_data)

        cluster_id = find_cluster(deploy, config['cluster']['name'])

        log_info("Found the cluster {} with id: {}.".format(config
['cluster']['name'], cluster_id))

        offline_cluster(deploy, cluster_id)

        delete_cluster(deploy, cluster_id)


def parseArgs():
    parser = argparse.ArgumentParser(description='Uses the ONTAP Select
Deploy API to delete a cluster')
```

```
    parser.add_argument('-d', '--deploy', required=True, type=str, help
='Hostname or IP address of Deploy server')
    parser.add_argument('-p', '--password', required=True, type=str, help
='Admin password of Deploy server')
    parser.add_argument('-c', '--config_file', required=True, type=str,
help='Filename of the cluster json config')
    return parser.parse_args()

if __name__ == '__main__':
    args = parseArgs()
    main(args)
```

공통 지원 모듈

모든 Python 스크립트는 단일 모듈에서 공통 Python 클래스를 사용합니다.

```
#!/usr/bin/env python
##-------------------------------------------------------------------
#
# File: deploy_requests.py
#
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#
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# that the software application product is distributed pursuant to terms
# no less restrictive than those set forth herein.
#
##-------------------------------------------------------------------

import json
import logging
import requests

requests.packages.urllib3.disable_warnings()

class DeployRequests(object):
    '''
    Wrapper class for requests that simplifies the ONTAP Select Deploy
    path creation and header manipulations for simpler code.
```

```
    '''

    def __init__(self, ip, admin_password):
        self.base_url = 'https://{}/api'.format(ip)
        self.auth = ('admin', admin_password)
        self.headers = {'Accept': 'application/json'}
        self.logger = logging.getLogger('deploy')

    def post(self, path, data, files=None, wait_for_job=False):
        if files:
            self.logger.debug('POST FILES:')
            response = requests.post(self.base_url + path,
                                     auth=self.auth, verify=False,
                                     files=files)
        else:
            self.logger.debug('POST DATA: %s', data)
            response = requests.post(self.base_url + path,
                                     auth=self.auth, verify=False,
                                     json=data,
                                     headers=self.headers)

        self.logger.debug('HEADERS: %s\nBODY: %s', self.filter_headers
(response), response.text)
        self.exit_on_errors(response)

        if wait_for_job and response.status_code == 202:
            self.wait_for_job(response.json())
        return response

    def patch(self, path, data, wait_for_job=False):
        self.logger.debug('PATCH DATA: %s', data)
        response = requests.patch(self.base_url + path,
                                  auth=self.auth, verify=False,
                                  json=data,
                                  headers=self.headers)
        self.logger.debug('HEADERS: %s\nBODY: %s', self.filter_headers
(response), response.text)
        self.exit_on_errors(response)

        if wait_for_job and response.status_code == 202:
            self.wait_for_job(response.json())
        return response

    def put(self, path, data, files=None, wait_for_job=False):
        if files:
            print('PUT FILES: {}'.format(data))
```

```
            response = requests.put(self.base_url + path,
                                    auth=self.auth, verify=False,
                                    data=data,
                                    files=files)
        else:
            self.logger.debug('PUT DATA:')
            response = requests.put(self.base_url + path,
                                    auth=self.auth, verify=False,
                                    json=data,
                                    headers=self.headers)

        self.logger.debug('HEADERS: %s\nBODY: %s', self.filter_headers
(response), response.text)
        self.exit_on_errors(response)

        if wait_for_job and response.status_code == 202:
            self.wait_for_job(response.json())
        return response

    def get(self, path):
        """ Get a resource object from the specified path """
        response = requests.get(self.base_url + path, auth=self.auth,
verify=False)
        self.logger.debug('HEADERS: %s\nBODY: %s', self.filter_headers
(response), response.text)
        self.exit_on_errors(response)
        return response

    def delete(self, path, wait_for_job=False):
        """ Delete's a resource from the specified path """
        response = requests.delete(self.base_url + path, auth=self.auth,
verify=False)
        self.logger.debug('HEADERS: %s\nBODY: %s', self.filter_headers
(response), response.text)
        self.exit_on_errors(response)

        if wait_for_job and response.status_code == 202:
            self.wait_for_job(response.json())
        return response

    def find_resource(self, path, name, value):
        ''' Returns the 'id' of the resource if it exists, otherwise None
'''
        resource = None
        response = self.get('{path}?{field}={value}'.format(
                            path=path, field=name, value=value))
```

```
        if response.status_code == 200 and response.json().get
('num_records') >= 1:
            resource = response.json().get('records')[0].get('id')
        return resource

    def get_num_records(self, path, query=None):
        ''' Returns the number of records found in a container, or None on
error '''
        resource = None
        query_opt = '?{}'.format(query) if query else ''
        response = self.get('{path}{query}'.format(path=path, query
=query_opt))
        if response.status_code == 200 :
            return response.json().get('num_records')
        return None

    def resource_exists(self, path, name, value):
        return self.find_resource(path, name, value) is not None

    def wait_for_job(self, response, poll_timeout=120):
        last_modified = response['job']['last_modified']
        job_id = response['job']['id']

        self.logger.info('Event: ' + response['job']['message'])

        while True:
            response = self.get('/jobs/{}?fields=state,message&'
                                'poll_timeout={}&last_modified=>={}'
.format(
                                    job_id, poll_timeout, last_modified))

            job_body = response.json().get('record', {})

            # Show interesting message updates
            message = job_body.get('message', '')
            self.logger.info('Event: ' + message)

            # Refresh the last modified time for the poll loop
            last_modified = job_body.get('last_modified')

            # Look for the final states
            state = job_body.get('state', 'unknown')
            if state in ['success', 'failure']:
                if state == 'failure':
                    self.logger.error('FAILED background job.\nJOB: %s',
job_body)
                    exit(1)   # End the script if a failure occurs
```

```
                break

    def exit_on_errors(self, response):
        if response.status_code >= 400:
            self.logger.error('FAILED request to URL: %s\nHEADERS: %s
\nRESPONSE BODY: %s',
                              response.request.url,
                              self.filter_headers(response),
                              response.text)
        response.raise_for_status()   # Displays the response error, and
exits the script

    @staticmethod
    def filter_headers(response):
        ''' Returns a filtered set of the response headers '''
        return {key: response.headers[key] for key in ['Location',
'request-id'] if key in response.headers}
```

클러스터 노드의 크기를 조정하는 스크립트입니다

다음 스크립트를 사용하여 ONTAP Select 클러스터에서 노드 크기를 조정할 수 있습니다.

```
#!/usr/bin/env python
##-------------------------------------------------------------------
#
# File: resize_nodes.py
#
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#
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# that the software application product is distributed pursuant to terms
# no less restrictive than those set forth herein.
#
##-------------------------------------------------------------------

import argparse
import logging
import sys
```

```
from deploy_requests import DeployRequests


def _parse_args():
    """ Parses the arguments provided on the command line when executing
this
        script and returns the resulting namespace. If all required
arguments
        are not provided, an error message indicating the mismatch is
printed and
        the script will exit.
    """

    parser = argparse.ArgumentParser(description=(
        'Uses the ONTAP Select Deploy API to resize the nodes in the
cluster.'
        ' For example, you might have a small (4 CPU, 16GB RAM per node) 2
node'
        ' cluster and wish to resize the cluster to medium (8 CPU, 64GB
RAM per'
        ' node). This script will take in the cluster details and then
perform'
        ' the operation and wait for it to complete.'
    ))
    parser.add_argument('--deploy', required=True, help=(
        'Hostname or IP of the ONTAP Select Deploy VM.'
    ))
    parser.add_argument('--deploy-password', required=True, help=(
        'The password for the ONTAP Select Deploy admin user.'
    ))
    parser.add_argument('--cluster', required=True, help=(
        'Hostname or IP of the cluster management interface.'
    ))
    parser.add_argument('--instance-type', required=True, help=(
        'The desired instance size of the nodes after the operation is
complete.'
    ))
    parser.add_argument('--ontap-password', required=True, help=(
        'The password for the ONTAP administrative user account.'
    ))
    parser.add_argument('--ontap-username', default='admin', help=(
        'The username for the ONTAP administrative user account. Default:
admin.'
    ))
    parser.add_argument('--nodes', nargs='+', metavar='NODE_NAME', help=(
        'A space separated list of node names for which the resize
```

```
operation'
        ' should be performed. The default is to apply the resize to all
nodes in'
        ' the cluster. If a list of nodes is provided, it must be provided
in HA'
        ' pairs. That is, in a 4 node cluster, nodes 1 and 2 (partners)
must be'
        ' resized in the same operation.'
    ))
    return parser.parse_args()


def _get_cluster(deploy, parsed_args):
    """ Locate the cluster using the arguments provided """

    cluster_id = deploy.find_resource('/clusters', 'ip', parsed_args
.cluster)
    if not cluster_id:
        return None
    return deploy.get('/clusters/%s?fields=nodes' % cluster_id).json
()['record']


def _get_request_body(parsed_args, cluster):
    """ Build the request body """

    changes = {'admin_password': parsed_args.ontap_password}

    # if provided, use the list of nodes given, else use all the nodes in
the cluster
    nodes = [node for node in cluster['nodes']]
    if parsed_args.nodes:
        nodes = [node for node in nodes if node['name'] in parsed_args
.nodes]

    changes['nodes'] = [
        {'instance_type': parsed_args.instance_type, 'id': node['id']} for
node in nodes]

    return changes


def main():
    """ Set up the resize operation by gathering the necessary data and
then send
        the request to the ONTAP Select Deploy server.
    """
```

```
    logging.basicConfig(
        format='[%(asctime)s] [%(levelname)5s] %(message)s', level=
logging.INFO,)

    logging.getLogger('requests.packages.urllib3').setLevel(logging
.WARNING)

    parsed_args = _parse_args()
    deploy = DeployRequests(parsed_args.deploy, parsed_args
.deploy_password)

    cluster = _get_cluster(deploy, parsed_args)
    if not cluster:
        deploy.logger.error(
            'Unable to find a cluster with a management IP of %s' %
parsed_args.cluster)
        return 1

    changes = _get_request_body(parsed_args, cluster)
    deploy.patch('/clusters/%s' % cluster['id'], changes, wait_for_job
=True)

if __name__ == '__main__':
    sys.exit(main())
```

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